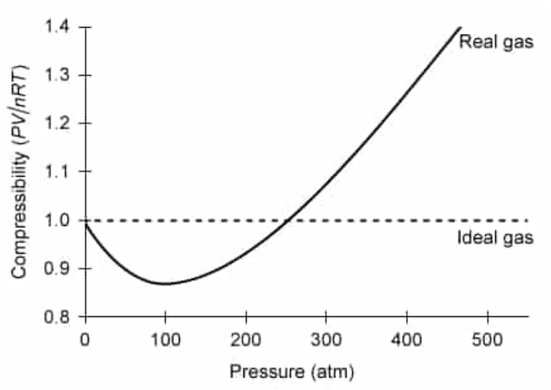
The compressibility of a real gas and an ideal gas at a temperature of 200 K is shown as a function of pressure in the graph below.

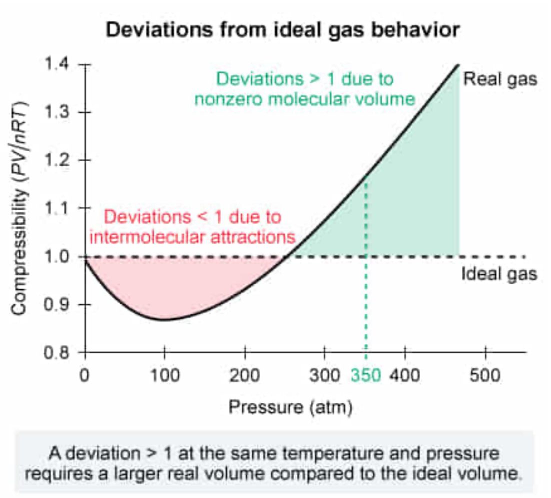


Real gases exhibit nonideal behavior because molecules occupy a finite molecular volume and intermolecular attractions present between molecules draws them closer together. Compared to the volume of the ideal gas at 350 atm, the volume of an equimolar amount of the real gas under identical conditions is:

- A. the same, because the pressure and temperature are the same for both gases. [7%]
- B. smaller, because the number of moles of each gas is the same but their molar masses are different. [10%]
- C. larger, because the combined volume of gas molecules is significant relative to the volume of the container. [40%]
- D. larger, because the intermolecular interactions between gas molecules are significant. [41%]

Correct
41% answered correctly

Explanation:



An **ideal gas** is a hypothetical gas with well-behaved (idealized) characteristics used as a **simplified model** to approximate the behavior of **real gases** according to the ideal gas law:

$$PV = nRT$$

The ratio of these gas law terms gives the gas compressibility:

$$\text{Compressibility} = \frac{PV}{nRT}$$

For 1 mole of an ideal gas, the compressibility ratio equals 1 at all pressures. However, for real gases under very low temperatures or high pressures, this ratio may be higher or lower than 1 because real gases exhibit characteristics that deviate from the ideal gas model.

A real gas with compressibility < 1 indicates **attractions between molecules**. These attractions decrease the force of the molecular collisions with the container wall, which gives a real gas a lower pressure and a **lower compressibility** than an ideal gas.

A real gas with compressibility > 1 indicates effects from molecular volume. Although the molecular volume of a real gas is very small, it is not zero. The **nonzero molecular volume** of a real gas diminishes the available container space through which the gas can move. This adds a small amount of non-compressible volume and gives a real gas a higher volume and **higher compressibility** than an ideal gas.

Therefore, in this question, the real gas has a larger volume than the ideal gas at 350 atm because the combined volume of gas molecules is significant compared to the volume of the container.

(Choice A) Because the two lines on the graph diverge as pressure increases, the volume of the real gas must be greater than the volume of the ideal gas, even if the pressure and temperature are the same.

(Choice B) Under the same *P* and *T* as the ideal gas, an equal number of moles of the real gas must have a *larger* actual *V* to achieve the higher compressibility seen on the graph. Additionally, Avogadro's law states that volume depends on the *moles* of the gas (not its molar mass).

(Choice D) Intermolecular interactions generally cause a real gas to have a *smaller* volume than an ideal gas at the same pressure.

Educational objective:
Deviations from ideal gas behavior occur in real gases when the gas molecules with a non-negligible volume have significant intermolecular interactions and occupy a system at high pressure and low temperature.

Time spent:0 sec
Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws